# Bivariate statistics

Lecture and worked examples

# Reference course

A line of best fit $y = ax + b$ gives estimates, not exact equalities. The coefficient $a$ describes the estimated average change in $y$ when $x$ increases by one unit. Interpolation stays within the range of observed values; extrapolation goes outside that range and must be presented with caution.

## Reference course

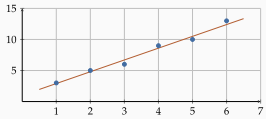
To use $y = ax + b$: 1) replace $x$ by the requested value to estimate $y$; 2) solve $ax + b = y_0$ to estimate $x$; 3) round according to the context; 4) state whether it is interpolation or extrapolation; 5) remind that the result depends on the model.

# Scatter plot

- Construct a scatter plot.
- Choose the scales.
- Identify the variables.

# Scatter plot

The table $(x_i, y_i)$ gives the advertising spending and the sales over six months:
$(1, 12), (2, 15), (3, 17), (4, 22), (5, 24), (6, 28)$. Choose scales, draw the scatter plot and plot the mean point.



# Scatter plot

A scatter plot represents the pairs $(x_i, y_i)$. The axes, units and scales must be indicated; each pair gives a single point.

# Scatter plot

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

# Scatter plot

**One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

**Two-hour plan:** add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

**Differentiation:** give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

# Reading and interpreting a scatter plot

- Describe a trend.
- Spot spread and outliers.
- Avoid unwarranted causality.

# Reading and interpreting a scatter plot

Compare three scatter plots: a tight increasing trend, a scattered decreasing trend, no trend. For each, write two sentences without using the word “cause”.

# Reading and interpreting a scatter plot

Describing a scatter plot means commenting on its direction, shape, spread and outliers, without claiming a causality that has not been established.

# Reading and interpreting a scatter plot

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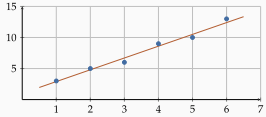
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# Linear regression

- Obtain a linear regression.
- Interpret $a$ and $b$.
- Check with the mean point.

# Linear regression

With the data of session 31, obtain the linear regression with a calculator. Interpret the slope and check that the mean point lies approximately on the line.



# Linear regression

A line $y = ax + b$ fits a scatter plot. $a$ represents the estimated average change in $y$ for one extra unit of $x$.

# Linear regression

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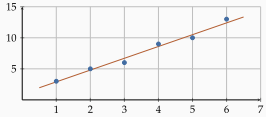
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# Interpolation

- Interpolate.
- Solve using the model.
- Express an estimate.

# Interpolation

Over the observed period $x \in [1, 12]$, the model is $y = 2.8x + 14$. Estimate $y$ for $x = 7.5$, then estimate $x$ when $y = 35$. Classify each estimate.



# Interpolation

An interpolation estimates a value inside the observed range. The result is an estimate that depends on the model.

# Interpolation

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# Interpolation

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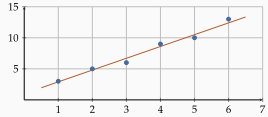
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# Extrapolation

- Extrapolate with caution.
- Assess plausibility.
- Present a limitation of the model.

# Extrapolation

The same model $y = 2.8x + 14$, fitted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it may be unreliable.



# Extrapolation

An extrapolation estimates outside the observed range. The further away we go, the more caution is needed.

# Extrapolation

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# Extrapolation

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# Change of variable

- Carry out a change of variable.
- Fit the new data.
- Return to the original variable.

# Change of variable

The data appear to be exponential. We set $z = \log(y)$ and the fit gives $z = 0.08x + 1.2$. Express $y$ as a function of $x$, then calculate $y$ for $x = 5$.

# Change of variable

A change of variable can make a scatter plot linear. After fitting in the new variable, we must return to the original variable.

# Change of variable

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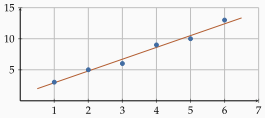
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# Choosing the model and fitting

- Compare models.
- Use residuals.
- Define a range of use.

# Choosing the model and fitting

Two models give total absolute residuals of 18.4 and 11.7 over the observed range. The second, however, shows unrealistic growth beyond $x = 20$. Choose a model to interpolate at $x = 12$, then discuss extrapolation at $x = 50$.



# Choosing the model and fitting

We choose a model according to the shape of the scatter plot, the size of the residuals and the range of use. A better visual fit does not allow unlimited extrapolation.

## Choosing the model and fitting

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# Choosing the model and fitting

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# Review and assessment

- Choose and use a regression.
- Write a statistical conclusion.
- Analyse an error.

## Review and assessment

Starting from a scatter plot supplied by the teacher, produce a complete answer: choice of model, equation, estimate, type of estimate, unit and caveat. Swap papers and use a six-criterion grid.

# Review and assessment

A complete statistical answer contains the model, the calculation, the rounding, the unit, the type of estimate and a caveat on validity.

# Review and assessment

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